Nepal College of Information Technology
Assessment Exam 2083

Program: BE CE VI

F.M : 100

P.M : 45

Subject: Machine Learning

✔ *Candidates are requested to give their answer as far as practicable in their own words.*
✔ *The figure in the margin indicates the full marks*
✔ *Attempt ALL question*

1) a) What is machine learning? Explain the working of machine learning with a clean diagram. [7]
b) Explain bias variance tradeoff. Explain different types of regularization techniques. [8]

2. a) What is supervised learning? Explain the working of ID3 algorithm for Decision Tree Classifier. [7]
b) Compare between hard marginal SVM and soft marginal SVM. Explain the working principle of linear SVM. [7]

3. a) What is unsupervised learning? For a given range of dataset (2,10),(2,5), (8,4), (5,8), (7,5), (6,4), (1,2), (4,9), (3,8), (6,12), create 3 clusters using the K-means clustering method. [8]
b) What is dimensionality reduction? Explain what is PCA and the steps involved in PCA. [7]
Or, You are given the following transaction database of a grocery store:
T1:{ Milk, Bread, Butter}
T2:{Milk, Bread}
T3:{Bread,Butter}
T4:{Milk,Butter}
T5:{Milk,Bread,Butter}
Assume:
Minimum support threshold: 60%
Minimum confidence threshold: 70%

4. a) Explain density based clustering. For the given data points identify density based clusters based on the given data and identify noises from it. Use ε=0.5 and minPts = 3 [8]
D={(1,1),(1.2,1.1),(0.8,1.0),(1.1,0.9),(1.3,1.2),(8,8),(8.2,7.9), (7.8,8.1),(20,1)}
b) What is a perceptron? Differentiate between single layer perceptron and multi layer perceptron with examples. [7]

5. a) Explain CNN architecture? Explain its architecture and how does CNN extract features from the image? [7]
b) What is RNN? Explain the disadvantages of RNN and how LSTM overcomes it? [8]

6. a) What is a confusion matrix? How does f1 score differ from accuracy model evaluation. How does ROC help in evaluating the binary classification model? [7]
b) Explain the concept of training, validation and testing data. How does K-fold cross validation work? [8]

7. Write short notes on: (Any Two) [10]
a) Hyperparameter Tuning
b) Reinforcement Learning
c) Loss Functions

compiled by Niraj Bhatta

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test I

Date: 2083/03/4			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	VI	1.5 hrs	

Subject: - Machine Learning

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt <u>All</u> questions.*
- ✓ *The figures in the margin indicate <u>Full Marks</u>.*
- ✓ *Assume suitable data if necessary.*

1. What is Machine Learning? Discuss it types along with their respective use cases. [8]
2. Explain the Machine Learning Workflow. Highlight the significance of feature scaling. [7]
3. Discuss the significance of Bias Variance tradeoff. How is it affected by regularization? Explain Ridge and Lasso Regression with their mathematical expressions. [8]
4. Describe how Logistic Regression technique can be used for binary and multi-class classification. [7]
5. The following data shows the relationship between a dependent variable Y and two independent variables X_1 and X_2. Assuming a multiple linear regression model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$, perform the following tasks. [8]

Observation	(Y)	(X_1)	(X_2)
1	6	1	2
2	8	2	1
3	10	3	4
4	12	4	3

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a. Write the model in matrix form $Y = X\beta$
b. Compute $X^T X$ and $X^T Y$
c. Estimate $\beta = (X^T X)^{-1} X^T Y$
d. Compute the value of Y for new inputs with $X_1 = 2$ and $X_1 = 3$.

OR

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Why do we require Polynomial Regression? Briefly explain the assumptions of Linear Regression.

6. What is SVM? Explain the workings of SVM Classifier. Discuss the importance of Kernel functions in SVM. [7]

OR

The following dataset is given. A new instance $Q = (3,4)$ is observed. Classify Q using k=3

ID	X_1	X_2	Class
1	1	2	A
2	2	3	A
3	3	1	B
4	6	5	B
5	7	7	B

7. Write Short Notes on: (***any one***) [5]
   a) Ethical Principles in Machine Learning
   b) Support Vector Regression
   c) Curse of Dimensionality

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Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Internal Examination

Level: Bachelor
Programme: B.E(Computer)
Year/Part: III/II

Full Marks: 60
Pass Marks: 27
Time: 2 hrs

Subject: - Machine Learning

- ✓ *Candidates are required to give their answers in their own words as far as possible.*
- ✓ *Attempt all question*

1. a) What is SVM? How does the SVM classifier work? Also explain the importance of kernel functions in SVM. (7)

b) Explain bias variance tradeoff? Explain the different types of regularization techniques. (7)

2. a) A medical clinic uses KNN to categorize patients. The features are Age (normalized) and Heart Rate (normalized). The target class is Risk Level (0 = Low, 1 = High). (8)

Patient	Age(X1)	Heart Rate(X2)	Risk Level(Y)
P1	0.25	0.4	0
P2	0.35	0.55	0
P3	0.7	0.75	1
P4	0.8	0.9	1
P5	0.45	0.3	0

OR

For the given data points, calculate the distance matrix and create a dendrogram using single linkage technique.

Point	P1	P2	P3	P4	P5
X	1	1	0	6	7
Y	4	2	4	1	0

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b) What is the role of Principal Component Analysis (PCA) in unsupervised learning? Explain the working mechanism of PCA. (7)

3. a) How multilayer perceptron uses the backpropagation and gradient descent in neural network training? Explain with an example. (7)

b) Describe the components of a Convolutional Neural Network (CNN) and explain their functions. (7)

OR

Explain the architecture and functionality of Long Short-Term Memory (LSTM) networks. How do they overcome the vanishing gradient problem?

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4. a) What are model validation techniques? Compare hold-out and k-fold cross-validation. (7)

b) Pokhara University recently developed a machine learning model to predict PASSED and FAILED students. While evaluating the model with a dataset of 1000 records, the system correctly predicted 480 students as PASSED and 380 students as FAILED students. However, 120 students were misclassified as FAILED students.

i. Create a confusion matrix. (7)

ii. Compute accuracy, precision, recall, F1 score, sensitivity and specificity. (4)

c) Write short notes on: (Any one)

i. DBSCAN

ii. Hyperparameter Tuning

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Pokhara University

Everest Engineering College

Final Internal Assessment
Spring- 2026

Level: Bachelor — **F.M.** 100
Program: BE CMP(6th Semester) — **P.M.** 45
Faculty: Science & Technology — **Time:** 3hrs
Subject: Machine Learning

Attempt all the questions.

1 a) Define and explain different types of Machine learning with an example. 7

b) Describe the SVM in classification with hyperplane, support vector and kernels. 8

2 a) Build the Multiple Linear Regression model for the given dataset. 8

y	X_1	X_2
140	60	22
155	62	25
159	67	24
179	70	20
192	71	15
200	72	14
212	75	14
215	78	11

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OR

Explain the regularization techniques in regression model with its mathematical details. 7

b) Explain the Bagging and random forest with its practical application.

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3 a) Define clustering and list out the use of clustering in ML. explain K means Clustering algorithm. 8

b) Apply the DBSCAN algorithm to the given dataset P1(3,7), P2(4,6), P3(5,5), P4(6,4), P5(7,3), P6(6,2), P7(7,2), P8(8,4). P9(3,3), P10(2,6), P11(3,5), P12(2,4). Create the clusters with 4 minpts and value of epsilon is 1.9. *[Use the attached table for distance between data points]* 7

4 a) What are the merits and demerits of dimensionality reduction? Explain any two techniques for dimensionality reduction. 8

b) What are the basic components of ANN? Describe on the basis of Feedforward Neural Network. 7

5 a) List the application of CNN. Also, explain Convolution operation in CNN in details. 8

b) Why do we need RNN even though CNNs are powerful? Explain RNN with its architectural diagram. 7

OR

What is perceptron? Explain single layer, multilayer and backpropagation in perceptron.

6 a) Why loss calculation is important in Neural network. Explain the cross-entropy loss. 7

b) Explain Cross validation and Hyperparameter tuning in model Evaluation and Validation. 8

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7 Write short notes on: (Any two) 2*5=10

a) Model Evaluation Matrix in Classification

b) ML Pipeline

c) Loss Functions in Training Neural Network

Distance table for data points

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	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.0	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

Best Wishes

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Pokhara Engineering College

Final Assessment

Program :Computer | ***FM: 100***

Level : Bachelor | ***PM: 45***

Year/Part : III / II | **Subject:** Machine Learning | ***Time: 3 hrs***

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks. ***Neat and clean writing are extra credited.***

Attempt all the questions.

1. a) Justify why Machine Learning is preferred over traditional programming in solving real-world problems. Support you answer with suitable examples. 7

   b) Given below simple dataset, determine the regression equation $Y = a + bX$ using the Least Squares Method. Hence predict the sales when the advertising expenditure is 10 units. 8

X (Advertising Expenditure)	Y (Sales)
2	30
4	45
6	55
8	70

2. a) Explain the concept of overfitting and underfitting. Compare Ridge and Lasso regression, highlighting their advantages and applications. 7

   b) "The choice of the kernel function significantly affects the performance of an SVM classifier." Justify this statement with suitable examples. 8

OR

Explain the K-Nearest Neighbor (KNN) classification algorithm. Discuss how the choice of the value of K affects the performance of the classifier, and illustrate your answer with a suitable example.

3. a) Consider the following two-dimensional dataset. Assume ε (Eps)=2 and MinPts=2. 7

   i) Compute the ε-neighborhood of each point.

   ii) Identify the core points, border points, and noise points.

   iii) Determine the clusters formed by the DBSCAN algorithm.

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Points	Coordinates (x,y)
P1	(1, 1)
P2	(2, 1)
P3	(2, 2)
P4	(8, 8)
P5	(8, 9)
P6	(25,25)

   b) Explain how FP-Growth algorithm overcomes the limitation of Apriori algorithm. Assume minimum support count=2 for below transactions, compute Frequent 2-itemsets and confidence for rule {2}→{3}. 8

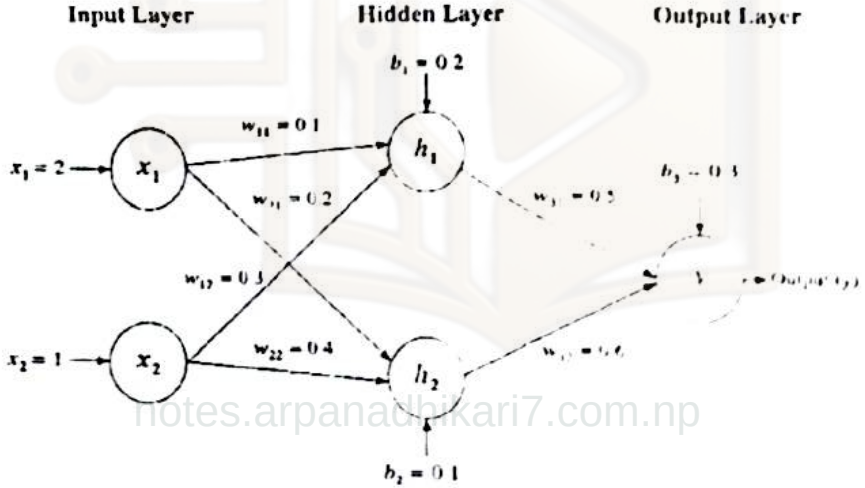
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TID	Item
501	3,
502	2,
503	1,
504	1,2,3,6

4. a) What is dimensionality reduction? Why is it required in Machine Learning? Explain how PCA is used for dimensionality reduction and compare it with LDA 7

b) Given the feedforward neural network shown in the figure, compute the outputs of the hidden layer and the output layer using the Sigmoid activation function. If the target output is 1, calculate the Mean Squared Error (MSE). Show all intermediate calculations. Is backpropagation required in this case? Justify your answer. 8



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5 a) Explain the Gradient Descent optimization algorithm. Describe how backpropagation uses gradient descent to minimize the loss function during neural network training. 7

b) Explain the architecture of a Long Short-Term Memory (LSTM) network. Discuss the functions of the input, forget, and output gates, and explain how they enable LSTM to capture long-term dependencies in sequential data. 8

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$b_2 = 0.1$

5 a) *Explain the Gradient Descent optimization algorithm. Describe how backpropagation uses gradient descent to minimize the loss function during neural network training.* 7

b) Explain the architecture of a Long Short-Term Memory (LSTM) network. Discuss the functions of the input, forget, and output gates, and explain how they enable LSTM to capture long-term dependencies in sequential data 8

OR

"Convolutional Neural Networks (CNNs) have become the preferred deep learning model for image recognition." Justify this statement by explaining the key components and advantages of CNNs.

6. a) A binary classification model was evaluated on a test dataset of 200 instances. The model correctly classified 80 positive instances and 90 negative instances. It incorrectly classified 20 positive instances as negative and 10 negative instances as positive. 7

i) Construct the confusion matrix.

ii) Calculate the Accuracy, Precision, Recall, and F1-Score of the model.

b) Explain the concept of hyperparameter tuning in Machine Learning. Compare Grid Search and Random Search methods for hyperparameter optimization. 8

7. Write short notes on: (any two) 10

a) Machine Learning workflow

b) K-fold crossvalidaton

c) Curse of Dimensionality

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NEPAL ENGILEERING COLLEGE

Batch: COMP-023 Semester: Spring Year: 2026
Program: BE Computer Full Marks: 100
Course: Machine Learning Pass Marks: 45
Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain the major types of Machine Learning techniques. For each technique, discuss its key characteristics and typical application areas. 3+4
   b) Describe the major challenges in machine learning, specifically focusing on data quality issues, computational complexity, and ethical considerations. 8
2. a) Given the dataset for a simple linear regression: n=10, $\sum x=100$, $\sum y=500$, $\sum xy=5400$, $\sum x^2=1200$. Compute the slope (b_1) and intercept (b_0). Find the predicted y when x=15, and if the actual y=58, compute the residual error. 7
   b) Explain the construction and pruning of Decision Trees with suitable illustrations. 8
3. a) An analyst wants to build a multiple linear regression model using the matrix method to predict a car's price. Explain the mathematical formulation to solve for the coefficient vector $\beta=(X^TX)^{-1}X^Ty$ and what the matrices represent. 7
   b) Differentiate Agglomerative and Divisive Hierarchical Clustering techniques. Explain, with the help of an algorithm, how the DIANA (Divisive Analysis) hierarchical clustering method works. 4+4

   OR

   Explain the vanishing gradient problem in recurrent neural networks (RNNs). How is the Long Short-Term Memory (LSTM) architecture designed to overcome this problem?
4. a) Explain the K-Means clustering algorithm and its limitations when applied to non-spherical data? Use the K(=2)-Means algorithm over a given set of 2D data points up to two iterations and show the resulting clusters, given the initial centroids. 3+4 4+4

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data_points = {"P1": (2, 10), "P2": (2, 5), "P3": (8, 4), "P4": (5, 8), "P5": (7, 5),
"P6": (6, 4), "P7": (1, 2), "P8": (4, 9)}
initial_centroids = {"C1": (2, 10), "C2": (5, 8)}

b) Explain the concept of a perceptron and how a multilayer perceptron overcomes the limitations of a single-layer perceptron discuss.

5. a) Describe the architecture of a Feedforward Neural Network. Explain the steps involved in the Backpropagation algorithm. 3+4 3+5

b) Describe the standard architecture of a Convolutional Neural Network (CNN) with illustration. Explain the specific roles and functions of its core components: convolution layers, pooling layers, flattening layers, and fully connected (dense) layers.

6. a) What are model validation techniques? Compare hold-out (train-test split) and k-fold cross-validation. 7

b) From a logistic regression model for heart disease (1 = present, 0 = absent), the actual and predicted values are: True Positives = 80, False Negatives = 20, False Positives = 30, and True Negatives = 70. Calculate accuracy, precision, recall (sensitivity), specificity, and the F1-score. 8

7. Write short notes on: (Any two) 2×5

a) Association Rule Learning
b) Support Vector Regression (SVR)
c) Bias-Variance tradeoff
d) DBSCAN
e) LDA

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Universal Engineering & Science College

Affiliated to Pokhara University

Chakupat, Lalitpur

Level: Bachelor	Semester: 6th	Year : 2026
Programme : BE Computer	Time : 3 hours	Full Marks: 100
Subject: Machine Learning		Pass Marks: 45

Pre-Board Examination- 2083(Spring 2026)

Candidates are required to give their answers in their own words as practicable. The figures in the margin indicate full marks.

Attempt All the questions:

1. a) What is Machine Learning? Explain the types of machine learning with examples. 7

   b) Explain the concept of Performance, Experience, and Task in the context of Machine Learning. 8

2. a) Describe different data preprocessing techniques in Machine Learning. Why is data cleaning and feature engineering important for building accurate models? 7

   b) Explain the working principle of the Support Vector Machine (SVM). Discuss the role of kernel functions with suitable examples. 8

   OR,

   What is gradient descent? Explain the working of gradient descent when minimizing cost function of linear regression.

3. a) Compute the linear regression equation using the Least Squares Method for the following data and predict the output for X = 7. 7

(X)	(y)
2	18
4	30
6	42
8	56

   b) Write down the algorithm for k-means clustering. Cluster the given eight points into three clusters: A1(2,10), A2(2,5), A3(8,4), A4(5,8), A5(7,5), A6(6,4), A7(1,2), A8(4,9). 8

4. a) Describe the steps involved in performing PCA, including identifying principal components. 7

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b) What is the Agglomerative Clustering algorithm? Construct a dendrogram for the following data using simple linkage method: 8

Sample No.	X	y
S1	4	3
S2	1	4
S3	2	1
S4	3	8
S5	6	9
S6	5	1

5. a) Explain the Error Backpropagation Learning Algorithm with a suitable block diagram. Discuss the role of activation functions and learning parameters. 7

b) What is a Single Layer Perceptron? Explain the Perceptron Learning Algorithm. Discuss the limitations of a single-layer perceptron with respect to the XOR problem. How can this limitation be overcome? 8

OR,

What is overfitting in the context of neural network training? Explain the difference between training, validation, and testing sets. Describe methods to prevent overfitting, such as dropout and early stopping.

6. a) Describe the architecture of a Convolutional Neural Network (CNN). Explain the functions of convolution, pooling, activation, and fully connected layers. 7

b) Explain the different model evaluation metrics of classification. 8

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7. Write short notes on: **(Any two)** 2×5
   a) Machine learning workflow
   b) Types of Regularization
   c) K-fold cross validation

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor
Program: B.E. Computer
Course: **Machine Learning (New)**

Semester: **VI Spring**

Year : 2026
F.M. : 100
P. M. : 45
Time : 3 Hrs.

Candidates are required to give their answer in their own words as for as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) What is Machine Learning? Explain the different types of machine learning with suitable real-world examples for each. 7
   b) Explain the working flow of machine learning with neat diagram. 8

2. a) Explain polynomial regression and discuss how regularization techniques (Ridge and Lasso) help in reducing overfitting. 7
   b) Given the data points (X, Y): (1,2), (2,3), (3,5), (4,4), (5,6), compute the regression coefficients (slope and intercept) using the least squares method and predict Y when X = 6 8

3. a) For a Sunburn dataset given below, construct a decision tree using ID3. 8

Name	Hair	Height	Weight	Location	Class
Binita	Blonde	Average	Light	No	Yes
Sarita	Blonde	Tall	Average	Yes	No
Prativa	Brown	Short	Average	Yes	No
Sushila	Blonde	Short	Average	No	Yes
Rawan	Red	Average	Heavy	No	Yes
Iris	Brown	Tall	Heavy	No	No
Raj	Brown	Average	Heavy	No	No
Anju	Blonde	Short	Light	Yes	No

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b) For the given data points, compute the distance matrix and construct a dendrogram using the compute linkage technique:

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

4. a) What is Dimensionality Reduction? Explain the working mechanism of PCA for dimension reduction. 8

b) What is unsupervised learning? For a given range of dataset (2,10), (2,5), (8,4), (5,8), (7,5), (6,4), (1,2), (4,9), create 3 clusters using the K-means clustering method. 7

5. a) Explain the structure of a biological neuron and relate it to the architecture of an artificial neuron (perceptron). 7

OR

Discuss different activation functions used in neural networks (sigmoid, tanh, ReLU) along with their advantages and disadvantages. 8

b) Describe the components of a Convolutional Neural Network (CNN) and explain their functions.

OR

What is a Recurrent Neural Network (RNN)? Explain how GRU is different from LSTM.

6. a) Explain the importance of cross-validation in Machine Learning. Compare Leave-One-Out Cross Validation (LOOCV) and K-Fold Cross Validation. 7

b) A machine learning model was evaluated on a dataset of 800 records to predict SPAM and NOT SPAM emails. The model correctly predicted 350 emails as SPAM and 380 as NOT SPAM. However, 70 emails were misclassified (40 actual SPAM predicted as NOT SPAM, and 30 actuals NOT SPAM predicted as SPAM). 8

i. Construct the confusion matrix.

ii. Compute Accuracy, Precision, Recall, F1-score, Sensitivity, and Specificity.

7. Write short notes on: (Any two) 2×5

a) Activation Functions b) Hyperplane and Margin in SVM

c) Bias-Variance Tradeoff and Hyperparameter Tuning

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2083/04/13			
Level	BE	Full Marks	50
Programme	BEIT	Time	
Semester	VI	1.5 hrs	

Subject: - Machine Learning

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt <u>All</u> questions.*
- ✓ *The figures in the margin indicate <u>Full Marks</u>.*
- ✓ *Assume suitable data if necessary.*

1. a) What is Association Rule Learning? From the datasets given below, Find the frequent item sets and association rules using Apriori algorithm. Assume Minimum Support Count = 2 and Minimum Confidence = 0.7. [8]

Transaction	Items Purchased
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Milk, Bread, Butter
T6	Bread, Butter

b) Discuss the concept of Density Based Clustering. Write step-by-step procedure of DBSACN clustering method. [7]

OR

Apply the K-Means (K=3) clustering algorithm to the following data point only up to two iterations and show the clusters and their centroids. Data points: (2, 2), (3, 3), (3, 4), (8, 7), (8, 8), (25, 80), (24, 78), and (26, 82). Use initial centroids as (2, 2), (8, 7), and (25, 80).

2. a) Differentiate between Forward Propagation and Backpropagation. Discuss how weights are updated during training. Why are activation functions important? [8]
   b) Explain the architecture of a Convolutional Neural Network (CNN) with a neat diagram. Describe the functions of each layers of CNN. [7]

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3. a) What is the significance of Model Validation. Discuss various model validation techniques. [8]
   b) A medical diagnosis system produced the following results: True Positives (TP) = 95, False Positives (FP) = 25, True Negatives (TN) = 160, False Negatives (FN) = 20. Calculate: Accuracy, Precision, Recall (Sensitivity), and F1-Score [7]

4. Write Short Notes on: (***any one***) [5]
   a) Principal Component Analysis
   b) Cross-Entropy Loss
   c) Bias–Variance Tradeoff

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[7]

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UNITED TECHNICAL COLLEGE

Semester: Spring

Year : 2026
Full Marks : 100
Pass Marks : 45
Time : 3hrs.

Level: Bachelor
Programme: BE
Course: Machine Learning
Semester: 6th

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1 a. Explain the Evolution of Machine Learning and the concept of Active Learning. Discuss how Active Learning addresses the challenges of data labelling compared to Supervised Learning. 7

b. Discuss the challenges in machine learning, such as data quality issues, computational complexity, interpretability, and ethical considerations. Provide real-world examples where these challenges affect model performance. 8

2 a. An analyst wants to predict a car's price based on engine size (X_1) and car age (X_2). Given the following dataset: 7

Car	X_1	X_2	Price (Y)
1	1.8	6	18
2	2.0	5	20
3	2.2	4	23
4	2.4	3	25
5	2.6	2	28

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Fit a Multiple Linear Regression model using the matrix method.

b. Discuss Logistic Regression for binary and multi-class classification problems. For a Logistic Regression model, the weights are $w_1 = 0.6, w_2 = -0.4$, and the bias is $b = 0.2$. Calculate the linear combination z for an input vector $X = (2, 3)$. Apply the Sigmoid function to determine the probability that the instance belongs to the positive class 8

3 a. Discuss the ensemble methods: Bagging and Random Forests. 7

OR

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Use the dataset below and calculate the Gini or Entropy for the root node and each possible split (just for root node), showing step-by-step calculation.

ID	Income	Student	Credit Rating	Buys Laptop
1	High	No	Fair	No
2	High	No	Excellent	No
3	Medium	No	Fair	Yes
4	Low	Yes	Fair	Yes
5	Low	Yes	Excellent	Yes
6	Low	No	Excellent	No
7	Medium	Yes	Fair	Yes
8	High	Yes	Fair	Yes
9	Medium	Yes	Excellent	Yes
10	High	No	Fair	No
11	Medium	No	Excellent	Yes
12	Low	Yes	Fair	Yes

b. Describe the Principal Component Analysis (PCA) algorithm. Explain how it reduces dimensionality with an example dataset (e.g., 2D to 1D projection). 8

4 a. Provide a step-by-step procedure for implementing Hierarchical Clustering (Agglomerative) on a given dataset. 8

Customer	Annual Income (k$)	Spending Score
A	15	39
B	16	81
C	17	6
D	18	77
E	19	40

b. Describe the training process of a neural network using forward propagation, backpropagation, and gradient descent. 7

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5 a. Explain overfitting and underfitting. Describe L1, L2 regularization and dropout. 7

b. Explain the architecture and working of a Convolutional Neural Network (CNN) in detail. Describe each of the following components clearly with appropriate diagrams and examples: 8

i. Convolutional layers: filters, stride, padding

ii. Pooling layers: max pooling vs average pooling

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Explain the architecture of Gated Recurrent Units (GRU). Describe how the update gate and reset gate help overcome the vanishing gradient problem in sequence prediction.

6 a. A classification model produced the following results: Correctly predicted 480 Passed students. Correctly predicted 380 Failed students. Misclassified 120 Passed students as Failed. Misclassified 20 Failed students as Passed. Construct the Confusion Matrix and calculate: Accuracy, Precision, Recall and F1-Score. 8

b. Describe model validation techniques: K-Fold Cross-Validation. Also explain hyperparameter tuning using Grid Search and Random Search. 7

7 (Attempt Any One) 5

a. SVM

b. DBSCAN

c. Reinforcement Learning.

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GANDKI COLLEGE OF ENGINEERING AND SCIENCE

Level: Bachelor
Programme: BE CE VI
Course: **Machine Learning**

Semester: Fall

Time: 3 hrs.

Year: 2026
Full Marks: 100
Pass Marks: 45

The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What are the different types of Machine Learning. Explain with relevant examples [7]
   b) Explain, with a diagram, the different steps involved in Machine Learning Workflow. [8]
2. a) Discuss about the different SVM Kernel Types. Provide example for each of them [7]
   b) What is Regularization. Differentiate between Ridge and Lasso Regularization [8]
3. a) What is Multiple Linear Regression? Provide an interpretation of the coefficients for the regression equation ($y = 25 + 5X_1 + 2X_2$) used to predict exam scores, where X_1 represents the number of hours studied and X_2 represents the attendance value. [7]
   b) Consider the given transaction database.
   (i) Assume minimum support count = 2. Find all frequent 1-itemsets and frequent 2-itemsets. Show support counts.
   (ii) For the rule Diaper → Beer, compute Support, Confidence, and Lift.

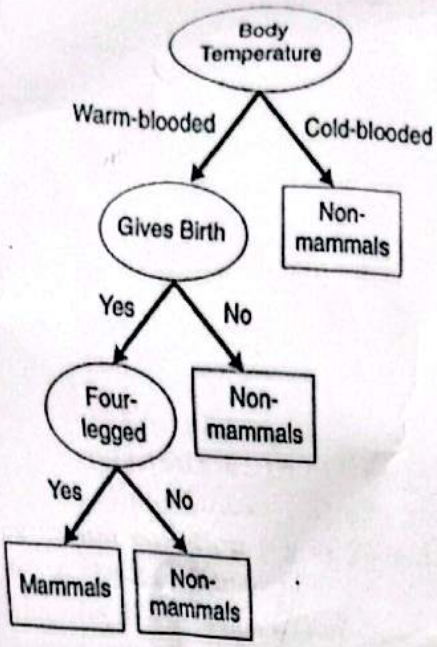
TID	Items
1	Rice, Milk
2	Rice, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Rice, Milk, Diaper, Beer
5	Rice, Milk, Diaper, Coke

4. a) Consider the decision tree and test dataset. Is the testing error rate the same as the training error rate? Calculate both and explain. [8]

Name	Body Temperature	Gives Birth	Four-legged	Hibernates	Class Label
Human	warm-blooded	yes	no	no	yes
Parrot	warm-blooded	no	no	no	no
Hippo	warm-blooded	yes	yes	no	yes
Shark	cold-blooded	yes	no	no	no
Turtle	cold-blooded	no	yes	no	no
Penguin	cold-blooded	no	no	no	no
Eel	cold-blooded	no	no	no	no
Dolphin	warm-blooded	yes	no	no	no
Spiny anteater	warm-blooded	no	yes	yes	yes
Gila monster	cold-blooded	no	yes	yes	no

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b) Consider the given data and perform DBSCAN clustering with $\varepsilon = 2$ and MinPts $= 2$. [7]

ID	A	B	C	D	E	F	G	H
x	2	2	8	5	7	6	1	4
y	10	5	4	8	5	4	2	9

5. a) What is Hierarchical Clustering. Tabulate the difference between Agglomerative and Divisive clustering. [8]

b) What is gradient descent? Show the analytical calculation for one weight update iteration when the objective loss function is $L(W) = W^2$, the starting weight is $W = 4$, and the step size is $\eta = 0.1$. [7]

6. a) Consider a single neuron with input vector $x = [2, 3]$, weight vector $w = [0, 1]$, bias $b = 4$, and sigmoid activation function. [8]

i) Compute weighted sum.

ii) Compute the neuron's output y (with sigmoid activation function).

b) A machine learning team trained a binary classification model to detect whether an email is spam or not. [7]

	Predicted Positive	Predicted Negative
Actual Positive	75	15
Actual Negative	10	100

Based on the provided confusion matrix, calculate the overall accuracy , precision, recall, and F1-score for the classification model.

7. Write short notes on **any two**: [8] [5 × 2]

a) Batch Normalization

b) Structured vs. unstructured data

c) PCA

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POKHARA UNIVERSITY
FACULTY OF SCIENCE & TECHNOLOGY

SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2026 Spring		
Level	B.E.	F M	100
Program	Computer	PM	45
Year/ Part	III/II	Time	3 Hrs

Subject: Machine Learning

Candidates are required to give answers in their own words as far as practicable. The figure in the margin indicates full marks. Assume suitable data if necessary. Attempt all the questions.

1. a) Explain the Machine Learning Workflow in detail covering Problem Definition, Data Collection and Pre-processing, Model Selection, Model Evaluation, and Model Deployment. Also discuss any two major challenges faced in Machine Learning. 7

   b) What are Regularization Techniques in Machine Learning? Explain Ridge Regression and Lasso Regression with their mathematical formulations. How does the Bias-Variance trade-off relate to regularization? 8

2. a) Consider the following dataset containing information about six students. The target variable Pass indicates whether a student passes the examination. 8

Student	Study Hours	Attendance	Pass
S1	High	High	Yes
S2	High	Low	Yes
S3	Low	High	Yes
S4	Low	Low	No
S5	Medium	High	Yes
S6	Medium	Low	No

Using the Entropy and Information Gain approach:

i. Calculate the entropy of the target variable Pass.
ii. Calculate the Information Gain for the attributes Study Hours and Attendance, and determine which attribute should be selected as the root node of the Decision Tree.

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b) Perform clustering using Agglomerative algorithm for the following Points: A(1, 2), B(1, 4), C(1, 0), D(4, 2), E(4, 4), F(4, 0) 7

OR

Apply the DBSCAN clustering algorithm for six data points in a two-dimensional space P1(1, 1), P2(1, 3), P3(2, 1), P4(2, 4), P5(5, 8), P6 (4, 7) using: Epsilon (ε) = 1.5, Minimum

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Points (MinPts) = 3. Calculate the Euclidean distance between the relevant points and identify the ε-neighborhood of each point.

3. a) A supermarket wants to discover relationships among products frequently purchased together. Explain how Association Rule Mining can be applied to this problem. Describe how the Apriori Algorithm abd FP-Growth Algorithm can be used to discover useful purchasing patterns. 8

b) What is a Convolutional Neural Network (CNN)? Explain its architecture and applications of CNN in image processing and computer vision. 7

4. a) Explain Artificial Neural Networks in detail. Describe the architecture of a Multilayer Perceptron (MLP). Explain Forward Propagation with mathematical formulations. Given: 10

- Input: X = [0.5, 0.3]
- Weights W = [0.4, 0.2, 0.3, 0.5]
- Bias b1 = [0.1]
- Activation function: Sigmoid $\sigma(x) = 1/(1+e^{-x})$

Calculate the output of the hidden layer after forward propagation.

b) Describe the concept of Backpropagation and Gradient Descent. 5

5. a) A Feedforward Neural Network processes each input independently, whereas an RNN is designed to process sequential data. Explain how an FNN can be transformed into an RNN and describe the complete working of the resulting RNN using an unrolled architecture. 7

b) Explain the Vanishing Gradient Problem in RNNs. Describe how the gating mechanisms of LSTM and GRU help in learning long-term dependencies and overcoming the limitations of traditional RNNs. 8

6. a) Why model validation techniques are required in machine learning? Explain the concept of train-test split. Discuss k-fold cross validation with an example how it works. 7

b) What is hyperparameter tuning? Explain Grid Search and Random Search with suitable diagrams and examples. 8

OR

Given confusion matrix shows the example of a binary classification model. Compute the accuracy, precision, recall and F1 score.

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n=165	Predicted: NO	Predicted: YES
Actual: NO	50	10
Actual: YES	5	100

7. Write short notes on: *(Any two)* 2 x 5

a) Support Vector Machine (SVM)

b) MAE & MSE

c) Overfitting and Underfitting in ML.

d) Applications of Machine Learning in Real Life

*** *Best of Luck* ***

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LUMBINI ENGINEERING MANAGEMENT & SCIENCE COLLEGE

(Affiliated to Pokhara University)
Tilottama -7 Bhalwari, Rupandehi

Final Term Examination - 2083

Programme : BE (Computer)
Semester : VI (Sixth)
Course : Machine Learning (ML)

Time : 3 Hrs
FM : 100
PM : 45

Candidates are requested to give their answer as far as practicable in their own words.
The figure in the margin indicates the full marks
Attempt ALL questions

1. a) Discuss any five major challenges of machine learning (e.g., data quality, computational complexity, interpretability, ethical issues) and briefly suggest how each can be mitigated. [7]

b) Differentiate between supervised learning, unsupervised learning, reinforcement learning, and active learning, with an example of a real-world application for each. [8]

2. a) What is regression? Explain simple linear regression and polynomial regression with their mathematical formulation. [7]

b) An analyst wants to build a multiple linear regression model to predict a house's price (in Lakhs) based on two factors: size of the house (in 100 sq. ft.) and age of the house (in years). Fit a multiple linear regression model using the matrix method. [8]

House	Size X_1 (100 sq.ft.)	Age X_2 (years)	Price (Lakhs)
1	15	10	45
2	18	8	52
3	12	15	38
4	20	5	60

3. a) Why is regularization important in machine learning? Explain Ridge regression and state the core differences between L1 and L2 regularization techniques. [7]

b) Given the following training dataset, use the K-Nearest Neighbors algorithm (with k = 3) and Euclidean distance to classify the test point (Age = 33, Income = 2). Show all distance calculations and the final decision. [8]

Point	Age	Income (in '000)	Class
A	25	40	No
B	35	60	Yes
C	45	80	Yes
D	20	20	No
E	52	90	Yes

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4. a) Use the K-Means (K = 2) algorithm on the following data points, up to two iterations, and show the resulting clusters. Choose Centroid 1: (2, 1) and Centroid 2: (9, 7) as the initial centroids. [8]

Data points: (1,1), (2,1), (2,3), (5,4), (6,5), (7,5), (9,7), (10,8)

b). What is the purpose of dimensionality reduction? Explain PCA (Principal Component Analysis) and the steps involved in performing it. How does LDA differ from PCA? [7]

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5.a) A dataset has 5 points in 2D: (2, 2), (2.5, 2.5), (8, 8), (8.4, 8.4), (8.6, 8.2). Apply DBSCAN with epsilon = 0.8 and MinPts = 2 (use Euclidean distance). Determine the clusters formed by DBSCAN, and identify any noise points. [8]

b) Explain the concept of a perceptron. Explain how the multilayer perceptron overcomes the limitations of a single-layer perceptron. [7]

6.a) Describe the architecture of a Convolutional Neural Network (CNN). Explain the role and working of the following key components with suitable diagrams: a) Convolutional Layer b) Pooling Layer c) Fully Connected Layer. [8]

b) A binary classifier gives the following confusion matrix for a dataset of 600 records. Calculate Accuracy, Precision, Recall, and F1-Score. Interpret the results. [7]

	Predicted Positive	Predicted Negative
Actual Positive	210	90
Actual Negative	60	240

7. Write Short Notes on: (any two) [5 x 2 = 10]

a. Random Forest
b. Logistic Regression
c. RMSE vs MAE

Best of Luck

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